Max Bothe

me@maxbothe.de
maxbothe.de
maxbothe
mathebox
Max Bothe

Skills

Specializations	Software Engineering, Data Analysis, Mobile Development, UI/UX Design
Qualifications	Project Management, Scrum (PSM I), Design Thinking

Programming

Advanced	Python, SQL, Swift, iOS Development
Intermediate	Ruby, Backend Development
Basic	Frontend Development

Other

Interests	Adobe Illustrator, Adobe Photoshop
Languages	German (native), English (fluent)

Experience

Since Jul 2017 **Senior Software Engineer**, *openHPI, Hasso Plattner Institute*, Potsdam, Germany

- Main responsible for the concept and development of the mobile applications for the HPI learning platform used by hundreds of thousands of mobile learners (`Swift, Kotlin`)
- Contributed to the design and development of *openHPI*, focusing on public APIs, accessibility and improvements to the user experience, impacting millions of users and key partners such as SAP and WHO (`Ruby on Rails, HTML5, JS, JSON:API`)
- Led the working group for user experience: conducting regular meetings as well as planning and implementing improvements (`Figma`)
- Collaborated with project partners and mentored new colleagues and student assistants

Oct 2015 – Jun 2017 **Teaching Assistant**, *Hasso Plattner Institute*, Potsdam, Germany

Mentored student teams in project management and taught design thinking methodologies for the course Global Team-based Product Innovation

Mar 2014 – Jun 2017 **Student Assistant**, *Hasso Plattner Institute*, Potsdam, Germany

Developed and designed an interactive tactic board for soccer analytics using geospatial data (`HTML5, JS, Django, SQL, SAP HANA`)

Apr 2013 – Sep 2013 **Software Developer Intern**, *SAP Research*, Belfast, United Kingdom

Developed a business process simulation that relies on BPMN models (`Spring, SQL`)

Aug 2012 – Feb 2013 **Mobile Developer Intern**, *Getemed AG*, Teltow, Germany

Developed an iOS application to configure longterm ECG recorders (`Objective-C, MFi technology by Apple`)

Education

Since Jul 2017 **Research Associate/Ph.D. Student**, *Hasso Plattner Institute*, Potsdam, Germany

- Researched the characteristic behavior and requirements of mobile learners in MOOCs
- Designed and implemented new features to support mobile learners in network-independent and short, interspersed learning activities
- Conducted several studies with thousand learners: formulating hypotheses and creating appropriate study designs (e.g., observational studies, randomized controlled trials, surveys)

(Cont.: Research Associate/Ph.D. Student)

- Analyzed large data sets with statistical methods to better understand learners' behavior and their learning achievements (`custom reporting tools of the platform, Python, SQL, ElasticSearch, ETL, pandas, seaborn, scipy, scikit-learn`)

Oct 2013 – Jun 2017 **Master of Science**, *Hasso Plattner Institute*, Potsdam, Germany
Majored in IT-Systems Engineering (Grade: 1.1)
Master's thesis: From MOOCs to Micro Learning: Optimizing mobile video-based learning

Oct 2014 – Jul 2015 **Joint Project on Global Team-based Product Innovation & Engineering**, *Stanford University*, California, USA
Participated successfully in the ME310 course series collaborating as a joint team-of-teams

Oct 2009 – Jul 2012 **Bachelor of Science**, *Hasso Plattner Institute*, Potsdam, Germany
Majored in IT-Systems Engineering (Grade: 1.8)
Bachelor's thesis: Implementation of an iOS application for configuration of longterm ECG recorders via Bluetooth

Projects (Excerpt)

Jun 2017 - Aug 2024 **Mobile Applications for the HPI Learning Platform**
Developed native mobile applications for iOS and Android for the HPI learning platform *openHPI*, which provide a streamlined design, optimized interactions and complementary features tailored to mobile use cases

Mar 2014 - Jun 2017 **Interactive Tactic Board for Soccer Analytics**
Developed an interactive tactics board based on geospatial data as a valuable tool for game analysis and training preparation, allowing coaches and professional athletes to analyze the specific way a game is set up and how team members perform in a game

Oct 2014 – Jul 2015 **Audi Ownership Experience in 2025**
Worked on a design challenge in cooperation with Stanford University, developing a prototype of the must have features of the Audi ownership experience in 2025 using design thinking methodologies

Mar 2012 – Feb 2013 **SEER 1000 iOS Application**, *Bachelor project & Internship*
Developed of an iOS application to configure longterm ECG recorders via Bluetooth

Publications (Excerpt)

Sep 2022 **Stay at Home and Learn: Did the COVID-19 Pandemic Influence Mobile Learning in MOOCs?**, *M. Bothe, Ch. Meinel. Proceedings of the 2022 IEEE Learning With MOOCS (LWMOOCS)*

Nov 2021 **Video Consumption with Mobile Applications in a Global Enterprise MOOC Context**, *M. Bothe, F. Schwerer, Ch. Meinel. Innovations in Learning and Technology for the Workplace and Higher Education (TLIC 2021)*

Jun 2021 **The Impact of Mobile Learning on Students' Self-Test Behavior in MOOCs**, *M. Bothe, Ch. Meinel. Eighth ACM Conference on Learning @ Scale 2021 (L@S '21)*

Sep 2020 **On the Potential of Automated Downloads for MOOC Content on Mobile Devices**, *M. Bothe, Ch. Meinel. Proceedings of the 2020 IEEE Learning With MOOCS (LWMOOCS)*

Apr 2020 **On the Acceptance and Effects of Recapping Self-Test Questions in MOOCs**, *M. Bothe, J. Renz, Ch. Meinel. Proceedings of the 2020 IEEE Global Engineering Education Conference (EDUCON)*

May 2019 **Applied Mobile-Assisted Seamless Learning Techniques in MOOCs**, *M. Bothe, Ch. Meinel. Digital Education: At the MOOC Crossroads Where the Interests of Academia and Business Converge*

Apr 2019 **From MOOCs to Micro Learning Activities**, *M. Bothe, J. Renz, T. Rohloff, Ch. Meinel. Proceedings of the 2019 IEEE Global Engineering Education Conference (EDUCON)*